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Inventor(s)	Viorel; Dorin Gheorghe et al.

Systems and methods for multiple repeater aggregation

Abstract

A method for multiple repeater aggregation includes (a) receiving, at a first node of a wireless communication network, wireless communication signals from a second node of the wireless communication network via a plurality of wireless communication paths, where each of the wireless communication paths includes a respective radio frequency (RF) repeater, and (b) selecting one of the plurality of wireless communication paths for use in transferring data between the first node of the wireless communication network and the second node of the wireless communication network, at least partially based on measured conditions of the plurality of wireless communication paths.

Inventors: Viorel; Dorin Gheorghe (Erie, CO), Sun; Ruoyu (Westminster, CO), Yi; Yunjung (Vienna, VA)

Applicant: Cable Television Laboratories, Inc. (Louisville, CO)

Family ID: 96950509

Assignee: Cable Television Laboratories, Inc. (Louisville, CO)

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Primary Examiner: Rampuria; Sharad

Attorney, Agent or Firm: PdZ Patent Law, PLLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority to U.S. Provisional Patent Application No. 63/279,581, filed on Nov. 15, 2021, which is incorporated herein by reference.

BACKGROUND

(1) Wireless communication networks transmit wireless communication signals between two or more wireless communication nodes. Examples of wireless communication nodes include, but are not limited to, wireless base stations, user equipment (UE) devices, and radio frequency (RF) repeaters. As such, performance of a wireless communication network is at least partially a function of quality of wireless communication links between wireless communication nodes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram of a wireless communication network including a system for multiple repeater aggregation (MRA), according to an embodiment.

- (2) FIG. 2 is a block diagram illustrating an embodiment of a method for MRA including selecting a single wireless communication path to transmit data between two wireless communication nodes.
- (3) FIG. 3 is a block diagram illustrating an embodiment of a method for MRA including combining copies of a common wireless communication signal.
- (4) FIG. 4 is a block diagram illustrating an embodiment of a method for MRA including joint transmission.
- (5) FIG. 5 is a graph illustrating one example allocation of communication resources in frequency and time across wireless communication paths of the FIG. 1 wireless communication network.
- (6) FIG. 6 is a block diagram of a wireless communication network illustrating an example of a method for load balancing, according to an embodiment.
- (7) FIG. 7 is a block diagram of an embodiment of the FIG. 6 wireless communication network configured to retransmit a data structure via an alternate path, in response to the data structure including errors.

DETAILED DESCRIPTION OF THE EMBODIMENTS

- (8) A distance between two wireless communication nodes may be too long to transmit robust wireless communication signals between the two wireless communication nodes. Additionally, an obstacle, such as by a tree, a building, or a natural feature, may block direct transmission of wireless communication signals between two wireless communication nodes. Radio frequency (RF) repeaters, and similar devices, may be deployed to at least partially overcome such problems by extending wireless communication network range and/or by providing alternate paths for wireless communication signals to travel between wireless communication nodes.
- (9) A standard RF repeater, sometimes referred to as a RF relay, simply receives wireless communication signals, amplifies the wireless communication signals, and forwards or “repeats” the wireless communication signals to another wireless communication node. For example, the Third Generation Partnership Project (3GPP) specifies a RF repeater in the fifth generation (5G) new radio (NR) Release 17 as a device which works on a RF layer and simply amplifies and forwards all wireless communication signals it receives. A standard RF repeater does not have wireless communication beam management capability, and a standard RF repeater is typically omnidirectional. While a standard RF repeater supports full-duplex communication, a standard RF repeater is not capable of distinguishing between downlink data and uplink data. Additionally, a standard RF repeater amplifies noise and interference along with wireless communication signals. As such, deployment of a standard RF repeater in a wireless communication network may or may not improve signal to interference plus noise ratio (SINR) in the wireless communication network, depending on the environment and the operating conditions of the wireless communication network.
- (10) A smart RF repeater (SR) has more functionality than a standard RF repeater. For example, the 3GPP 5G NR Release 18 specifies a smart RF repeater as a device which works on the RF layer of the user plane, and well as on the physical (PHY) and RF layers of the control plane. Release 18 also specifies that a smart repeater supports wireless communication beam management and that a smart repeater is capable of distinguishing between downlink data and uplink data.
- (11) Disclosed herein are new systems and methods which advance the state of the art of mesh wireless communication networks. Certain embodiments are operable with essentially any device which repeats wireless communication signals, including but not limited to, RF repeaters, RF relays, smart RF repeaters, and Integrated Access Backhaul (IAB) devices. Other embodiments leverage functionality of smart RF repeaters and similar devices.
- (12) The new systems and methods can be implemented in essentially any type of wireless communication network. For example, the new systems and methods could be embodied in 3GPP wireless communication networks, including, but not limited to, long term evolution (LTE) wireless communication networks, fifth generation (5G) wireless communication networks, sixth generation (6G) wireless communication networks, and future developed 3GPP wireless communication

networks. As another example, the new systems and methods could be embodied in non-3GPP wireless communication networks, including, but not limited to, Wi-Fi wireless communication networks, satellite wireless communication networks (e.g., using very low earth orbit (VLEO) satellites, low earth orbit (LEO) satellites, medium earth orbit (MEO) satellites, or geostationary equatorial orbit (GEO) satellites), Bluetooth wireless communication networks, long range (LoRa) wireless communication protocol, Zigbee wireless communication networks, Z-Wave wireless communication networks, etc. Furthermore, the new systems and methods could be embodied in wireless communication networks using licensed RF spectrum, unlicensed RF spectrum, or a combination of licensed and unlicensed RF spectrum.

(13) Multiple Repeater Aggregation

(14) Applicant has developed systems and methods for multiple repeater aggregation (MRA) which leverage multiple devices which repeat wireless communication signals, such as in a mesh wireless communication network. Certain embodiments are capable of improving wireless communication network coverage and reliability, increasing wireless communication channel capacity for downlink data and/or uplink data, and or extending wireless communication network coverage. For example, some embodiments enable a wireless communication network to maintain service to a wireless communication node even if some wireless communication signals to the wireless communication node are blocked by an obstacle, e.g., one wireless communication beam is blocked, thereby helping prevent shadowing loss. As another example, certain embodiments benefit from diversity gain, which mitigates small scale fading. As an additional example, particular embodiments are capable of serving a given wireless communication node via multiple channels with low correlation between the channels, or even with no correlation between the channels, which increases channel capacity according to Shannon theory.

(15) While the new systems and methods for MRA may achieve advantages in essentially any application, they may be especially beneficial in fixed wireless access (FWA) applications, vehicular applications, and aircraft applications. In particular, FWA applications may be particularly challenging due to FWA customer premises equipment (CPE) frequently being located indoors, as well as due FWA applications typically requiring more throughput than mobile applications. Use of the new MRA systems and methods may help improve coverage indoors and help achieve high throughput, which may make them particularly beneficial in FWA applications. Vehicular applications may also be particularly challenging due to vehicle movement, such as due to doppler shift resulting from vehicle movement. Use of the new MRA systems and methods may help mitigate such challenges. Furthermore, aircraft, such as drones, may be difficult to adequately serve using conventional wireless communication networks because conventional wireless communication networks are typically optimized for terrestrial users with down tilt in wireless base station antennas, causing coverage towards the sky to be poor. The new MRA systems and methods may be used in wireless communication networks to improve coverage towards the sky, which benefits aircraft applications.

(16) FIG. 1 is a block diagram of a wireless communication network **100** which includes an embodiment of the new systems for MRA. Wireless communication network **100** includes a wireless communication node **102**, a wireless communication node **104**, a RF repeater **106**, a RF repeater **108**, and a RF repeater **110**. It is understood, however, that wireless communication network **100** can include any quantity of wireless communication nodes and any quantity of RF repeaters, as long as wireless communication network **100** includes at least two wireless communication nodes and at least two RF repeaters. Particular embodiment of the new systems and methods for MRA do not require coordination among devices which repeat wireless communication signals. As such, in certain embodiments, RF repeaters **106**, **108**, and **110** are conventional RF repeaters or conventional RF relays. However, RF repeaters **106**, **108**, and **110** could be more sophisticated devices, such smart RF relays or IAB devices, as well as custom devices, without departing from the scope hereof.

(17) Each of wireless communication node **102** and wireless communication node **104** is configured to transmit and receive wireless communication signals. Wireless communication node **102** includes a wireless base station **112** and a MRA module **114**. In some embodiments, wireless base station **112** includes a 3GPP wireless base station, including, but not limited to, a NodeB, an eNode B, a gNodeB, or a future-developed 3GPP wireless base station. In some other embodiments, wireless base station **112** is a non-3GPP wireless base station, including, but not limited to, a Wi-Fi wireless base station, a satellite wireless base station, a Bluetooth wireless base station, a LoRa wireless base station, a Zigbee wireless base station, a Z-Wave wireless base station, etc. It should be noted that wireless base station **112** need not be a complete wireless base station. For example, in particular embodiments, wireless base station **112** is a remote radio head of a wireless base station. Additionally, wireless base station **112** could be replaced with a different type of device capable of transmitting and receiving wireless communication signals, such as a RF repeater or a user equipment (UE) device.

(18) Wireless communication node **104** includes a UE device **116** and a MRA module **118**. Although UE device **116** is depicted as being a mobile phone, UE device **106** could be another type of UE device. For example, UE device **106** could alternately be a computer, a set-top device, a data storage device, an Internet of Things (IoT) device, an entertainment device, a computer networking device, a modem, a smartwatch, a wearable device with wireless capability, a medical device, a security device, a monitoring device, or a wireless access device (including, for example, an eNB, a gNB, a Wi-Fi-based wireless access point, an IAB access point, a microcell, a picocell, a femtocell, a macrocell, a Wi-Fi-based application, a satellite communication device, etc.). Additionally, UE device **116** could be replaced with a different type of device capable of transmitting and receiving wireless communication signals, such as a RF repeater or a wireless base station.

(19) Wireless communication network **100** includes four wireless communication paths, i.e., wireless communication path **0**, wireless communication path **1**, wireless communication path **2**, and wireless communication path **3**, between wireless communication nodes **102** and **104**. Wireless communication path **0** is a direct path between wireless communication node **102** and wireless communication node **104** via a wireless channel Ch**0**. Accordingly, wireless communication path **0** does not include a RF repeater. Each of wireless communication paths **1**, **2**, and **3** includes a RF repeater. Specifically, wireless communication path **1** includes (i) a wireless channel Ch**11** between wireless communication node **102** and RF repeater **106** and (ii) a wireless channel Ch**12** between RF repeater **106** and wireless communication node **104**. Wireless communication path **2** includes (i) a wireless channel Ch**21** between wireless communication node **102** and RF repeater **108** and (ii) a wireless channel Ch**22** between RF repeater **108** and wireless communication node **104**. Wireless communication path **3** includes (i) a wireless channel Ch**31** between wireless communication node **102** and RF repeater **110** and (ii) a wireless channel Ch**32** between RF repeater **110** and wireless communication node **104**. Any of wireless communication paths **0**, **1**, **2**, and **3** could include a plurality of wireless communication beams, and each wireless communication beam could include a plurality of streams. Any one of wireless communication paths **0**, **1**, **2**, and **3** could carry downlink data, uplink data, or a combination of downlink data and uplink data.

(20) It is understood that the number of wireless communication paths between wireless communication nodes **102** and **104** will vary with the number of RF repeaters in wireless communication network **100**. Additionally, in some embodiments, one or more wireless communication paths between wireless communication nodes **102** and **104** may span two or more RF repeaters. Furthermore, a given wireless communication path between wireless communication nodes **102** and **104** may not be available due to, for example, the path being blocked by an obstacle.

(21) MRA module **114** and MRA module **118** collectively form an embodiment of the new systems for MRA. In some alternate embodiments, however, one of MRA module **114** and MRA module **118** is omitted, such that solely MRA module **114** or solely MRA module **118** forms the system for

MRA. Each of MRA modules **114** and **118** is implemented, for example, by a processor (not shown) executing instructions in the form of software and/or firmware stored in a data store (not shown). In some embodiment, MRA modules **114** and **118** are at least partially implemented by electronic circuitry (not shown). While MRA module **114** is depicted as being separate from wireless base station **112**, MRA module **114** may be partially or fully integrated with wireless base station **112**. For example, in certain embodiments, MRA module **114** is implemented by a processor of wireless base station **112** executing instructions in the form of software and/or firmware stored in a data store (e.g., a memory or a hard drive) of wireless base station **112**. Similarly, although MRA module **118** is depicted as being separate from UE device **116**, MRA module **118** may be partially or fully integrated with UE device **116**. For example, in certain embodiments, MRA module **118** is implemented by a processor of UE device **116** executing instructions in the form of software and/or firmware stored in a data store, e.g., a memory, of UE device **116**.

(22) The system for MRA of wireless communication network **100** is configured to execute one or more of the following methods for MRA, to promote high performance of wireless communication network **100**: (a) Method A-select of a single path, (b) Method B-combine copies of a signal, (c) Method C-joint transmission, (d) Method D-dynamic beam configuration, (c) Method E-beam direction, (f) Method F-multiple streams, (g) Method G-frequency/time resource aggregation, (h) Method H-fake base station detection, and (i) Method I-dynamic power control.

(23) Method A—Select a Single Path

(24) In method A, each RF repeater **106**, **108**, and **110** amplifies and repeats a received downlink or uplink wireless communication signal. Additionally, each RF repeater **106**, **108**, and **110** may receive the same copy of a downlink wireless communication signal or an uplink wireless communication signal sent by wireless base station **112** or UE device **116**, respectively. As such, there may be multiple wireless communication paths for the same copy of a wireless communication signal between wireless communication node **102** and wireless communication node **104**. However, these multiple wireless communication paths may not be equal. For example, one RF repeater **106**, **108**, and **110** may be blocked while another RF repeater **106**, **108**, and **110** may not be blocked. In method A, MRA module **114** and/or MRA module **118** select a single wireless communication path to transmit data between wireless communication node **102** and wireless communication node **104**, such as based on one or more measured conditions of the wireless communication paths, to help optimize wireless communication between the two nodes.

(25) In particular, one or more devices of wireless communication network **100** measure conditions of one or more wireless communication paths, or portions of wireless communication paths, of wireless communication network **100**, and the devices send data representing the measured conditions to MRA module **114** and/or MRA module **118**. For example, wireless base station **112** may measure conditions of wireless channels Ch**0**, Ch**11**, Ch**21**, and Ch**31** and send data representing the measured conditions to MRA module **114** and/or MRA module **118**. As another example, UE device **116** may measure conditions of wireless channels Ch**0**, Ch**12**, Ch**22**, and Ch**32** and send data representing the measured conditions to MRA module **114** and/or MRA module **118**. As additional example, RF repeater **106** may measure conditions of wireless channels Ch**11** and Ch**12** and send data representing the measured conditions to MRA module **114** and/or MRA module **118**. Examples of possible measured conditions of wireless communication paths include, but are not limited to, one or more of reference signal received power (RSRP) of each wireless communication path, reference signal received quality (RSRQ) of each wireless communication path, time advance of each wireless communication path, cell ID of each wireless communication path, wireless communication beam ID of each wireless communication path, percentage of utilization of each wireless communication path, etc.

(26) MRA module **114**, MRA module **118**, or the combination of MRA module **114** and MRA module **118**, select one wireless communication path to transmit data between wireless communication node **102** and wireless communication node **104** at least partially based on the data

representing the measured conditions. For example MRA module **114** and/or MRA module **118** may select whichever wireless communication path **1**, **2**, **3**, or **4** which has a highest RSRP, a highest RSRQ, a smallest time advance, a smallest percentage of channel utilization, etc. MRA module **114** and/or MRA module **118** then cause data to be routed between wireless communication node **102** and wireless communication node **104** solely via the one selected wireless communication path, such as by controlling wireless base station **112** and/or UE device **116** to route data accordingly. It should be noted that particular embodiments of Method A do not require coordination of RF repeater **106**, RF repeater **108**, or RF repeater **110**, thereby enabling the RF repeaters to be conventional RF repeaters in Method A.

(27) FIG. 2 is a block diagram of wireless communication network **100** illustrating one example of Method A. In this example, wireless communication paths **0**, **1**, and **2** are partially or completely blocked by a building **202**, a tree **204**, and a building **206**, respectively. MRA module **114** and/or MRA module **118** may analyze one or more measured conditions of wireless paths **0**, **1**, **2**, and **3** and determine that path **3** is the best path to transfer data between wireless communication node **102** and wireless communication node **104**, such as in response to wireless communication path **3** having a highest RSRP, a highest RSRQ, a smallest time advance, or a smallest percentage of wireless communication path. In response thereto, MRA module **114** and/or MRA module **118** cause wireless base station **112** and/or UE device **116** to route data between **102** and wireless communication node **104** solely via wireless communication path **1**, i.e., using channel Ch31, RF repeater **110**, and Ch32.

(28) Method B—Combine Copies of a Signal

(29) Referring again to FIG. 1, in method B, multiple copies of a wireless communication signal, i.e., either a downlink wireless communication signal or an uplink wireless communication signal, are transmitted between wireless base station **112** and UE device **116** via different respective wireless communication paths. In contrast with Method A, multiple copies of a wireless communication signal may be used by wireless base station **112** and UE device **116** in Method B. In particular, MRA module **114** and/or MRA module **118** cause wireless base station **112** to combine copies of an uplink wireless communication signal it receives via two or more of wireless communication paths **0**, **1**, **2**, and **3** to generate a combined wireless communication signal. Similarly, MRA module **114** and/or MRA module **118** cause UE device **116** to combine copies of a downlink wireless communication signal it receives via two or more of wireless communication paths **0**, **1**, **2**, and **3** to generate a combined wireless communication signal. Each combined wireless communication signal may have a greater amplitude than any one of its constituent copies of the wireless signal communication signal, which advantageously helps achieve a high signal to noise ratio. In some embodiments, the copies of the wireless communication signal are combined using a soft combining process. For example, wireless base station **112** may sum complex values of received copies of an uplink wireless communication signal to generate a combined uplink wireless communication signal, and UE device **116** may sum complex values of received copies of a downlink wireless communication signal to generate a combined downlink wireless communication signal. Certain embodiments of Method B do not require coordination of RF repeater **106**, RF repeater **108**, or RF repeater **110**, thereby enabling the RF repeaters to be conventional RF repeaters in Method B.

(30) Some embodiments of MRA module **114** and/or MRA module **118** are configured to cause wireless base station **112** and UE device **116** to combine only a predetermined quantity of received copies of a wireless communication signal, or to combine only received copies of a wireless communication signal that meet one or more predetermined criteria. For example, in one embodiment, MRA module **114** and/or MRA module **118** are configured to cause wireless base station **112** or UE device **116** to combine the two received copies of a wireless communications signal having largest RSRP values, largest RSRQ values, or smallest time advance values. As another example, in another embodiment, MRA module **114** and/or MRA module **118** are

configured to cause wireless base station **112** or UE device **116** to combine only copies of a wireless communication signal having RSRP values exceeding a threshold value, RSRQ values exceeding a threshold value, or time advance values that are below a threshold value.

(31) FIG. **3** is a block diagram of wireless communication network **100** illustrating one example of Method B. In this example, wireless communication paths **0** and **1** are blocked by a building **302** and a tree **304**, respectively. Consequently, wireless communication node **104** is capable of receiving copies of a downlink wireless communication signal solely via wireless communication paths **2** and **3**. Accordingly, UE device **116** receives copies **306** and **308** of a common downlink wireless communication signal, and UE device **116** combines these two copies to generate a combined downlink wireless communication signal **310**, as symbolically shown by text “**306+308=310**” in wireless communication node **104**. The magnitude of combined downlink wireless communication signal **310** is greater than the respective magnitudes of either of copies **306** and **308** of the downlink wireless communication signal. Therefore, Method B increases signal to noise ratio of downlink wireless communication signals processed by UE device **116**. UE device **116** could analogously receive and combine copies of an uplink wireless communication sent via wireless communication paths **2** and **3**, in the FIG. **3** example.

(32) Method C—Joint Transmission

(33) Referring again to FIG. **1**, in Method C. MRA module **114** and MRA module **118** cooperate such that data is transferred between wireless communication node **102** and wireless communication node **104** using joint transmission, such that different downlink data and/or uplink data are transferred via two or more of wireless communication paths **0**, **1**, **2**, and **3**. For example, in particular embodiments, MRA modules **114** and **118** are configured such that wireless base station **112** sends different downlink data to each of RF repeaters **106**, **108**, and **110** via respective wireless communication signals in wireless channels Ch**11**, Ch**21**, and Ch**31**. RF repeaters **106**, **108**, and **110** then forward the received downlink data to UE device **116** via respective wireless communication signals in each of wireless channels Ch**12**, Ch**22**, and Ch**32**, and UE device **116** aggregates the received wireless communication signals. Accordingly, UE device **116** receives downlink data from wireless base station **112** via multiple wireless communication paths. Similarly, in certain embodiments, MRA modules **114** and **118** are configured such that UE device **116** sends different uplink data to each of RF repeaters **106**, **108**, and **110** via respective wireless communication signals in each of wireless channels Ch**12**, Ch**22**, and Ch**32**. RF repeaters **106**, **108**, and **110** then forward the received uplink data to wireless base station **112** via respective wireless communication signals in each of wireless channels Ch**11**, Ch**21**, and Ch**31**, and wireless base station **112** aggregates the received wireless communication signals. Accordingly, wireless base station **112** receives different uplink from UE device **116** via multiple wireless communication paths. Direct path **0**, i.e., wireless channel Ch**0**, may be additionally be used in conjunction with one or more of wireless communication paths **1**, **2**, and **3** to further increase number of data transfer paths between wireless communication nodes **102** and **104**.

(34) Such ability to send different data via multiple wireless communication paths in Method C enables wireless communication network **100** to achieve a significantly larger throughput than would be feasible using conventional techniques. Additionally, the multiple wireless communication paths will typically be at different angles with respect to each of wireless base station **112** and UE device **116**, such that there may be small correlation, or even essentially no correlation between wireless communication paths **0**, **1**, **2**, and **3**, which further increases maximum theoretical throughput of wireless communication network **100**. Furthermore, use of multiple wireless communication paths for data transmission in wireless communication network **100** helps mitigate shadowing, small-scale fading, and Doppler shift. Furthermore, certain embodiments of Method C do not require coordination of RF repeater **106**, RF repeater **108**, or RF repeater **110**, thereby enabling the RF repeaters to be conventional RF repeaters.

(35) FIG. **4** is a block diagram of wireless communication network **100** illustrating one example of

Method C. In this example, wireless communication path 1 is blocked by a tree 402. However, wireless communication paths 0, 2, and 3 are available for data transmission between wireless communication nodes 102 and 104. Accordingly, MRA module 114 and MRA module 118 cooperate to provide for joint transmission of data between wireless base station 112 and UE device 116 using wireless communication paths 0, 2, and 3. For example MRA module 114 and MRA module 118 may cooperate such that wireless base station 112 sends different downlink wireless communication signals 404, 406, and 408 to UE device 116 in wireless communication via paths 0, 2, and 3. Specifically, in this example, wireless base station 112 sends downlink wireless communication signals 404 directly to UE device 116 using wireless communication path 0, and wireless base station 112 sends downlink wireless communication signals 406 and 408 via wireless channels Ch21 and Ch31 to RF repeater 108 and RF repeater 110, respectively. RF repeaters 108 and 110 then forward downlink wireless communication signals 406 and 408 to UE device 116, respectively, and UE device 116 aggregates received wireless communication signals 404, 406, and 408.

(36) Method D—Dynamic Beam Configuration

(37) Referring again to FIG. 1, in certain embodiments, each wireless each RF repeater 106, 108, and 110 may receive and transmit a plurality of beams. For example, wireless channel Ch11 between wireless base station 112 and RF repeater 106 may include a plurality of beams, and wireless channel Ch12 between RF repeater 106 and UE device 116 may also include a plurality of beams. As such, each of wireless communication paths 0, 1, 2, and 3 may include a plurality of possible beams by which wireless communication signals may be transmitted. In method D, MRA module 114 and/or MRA module 118 select a single wireless communication beam, or a single set of wireless communication beams, of a single wireless communication path 0, 1, 2, or 3 to transmit data between wireless communication node 102 and wireless communication node 104, such as based on one or more measured conditions of the wireless communication paths, to help optimize wireless communication between the two nodes. As such, method D is similar to method A, but method D further includes wireless communication beam coordination. The measured conditions of wireless communication paths in method D may include measured conditions of respective wireless communication beams of the wireless communication paths.

(38) In certain embodiments of Method D, UE device 116 may measure conditions of each wireless communication beam received from each RF repeater 106, 108, and 110, and UE device 116 may send data representing the measured conditions of the wireless communication beams to MRA module 114 and/or MRA module 118. Similarly, wireless base station 112 may measure conditions of each wireless communication beam received from each RF repeater 106, 108, and 110, and wireless base station 112 may send data representing the measured conditions of the wireless communication beams to MRA module 114 and/or MRA module 118. Such measurements are performed, for example, on a periodic basis, on an aperiodic basis, or in response to event. MRA module 114 and/or MRA module 118 may then select a single wireless communication beam, or a single set of wireless communication beams, for transmitting data between wireless communication nodes 102 and 104, based least in part on the data representing the measured conditions of the wireless communication beams. For example, MRA module 114 and/or MRA module 118 may select whichever wireless communication beam, or set of wireless communication beams, has a highest RSRP, a highest RSRQ, a smallest time advance, a smallest percentage of channel utilization, etc. MRA module 114 and/or MRA module 118 then cause data to be routed between wireless communication node 102 and wireless communication node 104 solely via the one selected wireless communication beam, or selected set of wireless communication beams, such as by controlling wireless base station 112 and/or UE device 116 to route data accordingly.

(39) In particular embodiments, wireless base station 112 and/or UE device 116 may not distinguish between wireless communication beams sent directly between the two devices and wireless communication beams sent between the two devices via one or more RF repeaters 106,

108, and **110**. Additionally, some embodiments of Method D do not require coordination of RF repeater **106**, RF repeater **108**, or RF repeater **110**, thereby enabling the RF repeaters to be conventional RF repeaters.

(40) Method E—Beam Direction

(41) Method E is an alternate embodiment of Method B further including wireless communication beam coordination. In Method E, multiple copies of a downlink wireless communication signal or an uplink wireless communication signal are sent by wireless base station **112** or UE device **116** via different respective wireless communication beams. For example, wireless base station **112** may send four copies of a common downlink wireless communication signal via one or more wireless communication beams along each of wireless communication paths **0**, **1**, **2**, and **3**. As another example, UE device **116** may send four copies of an uplink wireless communication signal via one or more wireless communication beams along each of wireless communication paths **0**, **1**, **2**, and **3**. Each RF repeater **106**, **108**, and **110** amplifies and forwards each wireless communication beam that it receives. As such, if all four wireless communication paths **0**, **1**, **2**, and **3** are available, UE device **116** will receive four copies of a common downlink wireless communication signal via four wireless communication beams. Similarly, if all four wireless communication paths **0**, **1**, **2**, and **3** are available, wireless base station **112** will receive four copies of a common uplink wireless communication signal via four wireless communication beams. MRA module **114** and/or MRA module **118** cause wireless base station **112** to combine copies of an uplink wireless communication signal it receives via two or more wireless communication beams. Additionally, MRA module **114** and/or MRA module **118** cause UE device **116** to combine copies of a downlink wireless communication signal it receives via two or more wireless communication beams.

(42) Method F—Multiple Streams

(43) In particular embodiments of wireless communication network **100**, wireless each RF repeater **106**, **108**, and **110** may receive and transmit a plurality of beams, and each beam may include a plurality of streams. Method F is similar to method C, but joint transmission is extended across multiple streams in each of two or more of wireless communication paths **0**, **1**, **2**, and **3**. For example, assume that each of wireless communication paths **0**, **1**, **2**, and **3** is available, and that each path supports two streams between wireless communication nodes **102** and **104**. In this example, there are eight routes between wireless communication nodes **102** and **104**, which significantly increases throughput of wireless communication network **100** compared to a scenario where joint transmission is not used. Additionally, in a manner analogous to that discussed above with respect to Method C, these multiple wireless communication paths enabled by Method F will typically be at different angles with respect to each of wireless base station **112** and UE device **116**, such that there may be small correlation, or even essentially no correlation between paths **0**, **1**, **2**, and **3**, which further increases maximum theoretical throughput of wireless communication network **100**. Furthermore, use of multiple wireless communication paths for data transmission in wireless communication network **100** helps mitigate shadowing, small-scale fading, and Doppler shift.

(44) Method G—Frequency/Time Resource Aggregation

(45) Methods A-F, discussed above, advantageously implement spatial diversity of communication resources to promote high performance of wireless communication network **100**. Any of methods A-F can be modified to further implement frequency and/or time diversity of communication resources among wireless communication paths **0**, **1**, **2**, and **3**, among wireless communication beams of a given wireless communication path, and/or among streams of a given wireless communication beam. Such implementation of frequency and/or time diversity in wireless communication network **100** may further promote high throughput and high reliability.

(46) FIG. 5 is a graph **500** of frequency versus time illustrating one example of how frequency and time diversity may be implemented by MRA module **114** and/or MRA module **118**. In the FIG. 5 example, a respective physical resource block (PRB) **502**, **504**, **506**, and **508** is allocated to each of

wireless communication path **0**, wireless communication path **1**, wireless communication path **2**, and wireless communication path **3**. Each PRB specifies a combination of frequency and time that may be used for the path. For example, wireless base station **112**, RF repeater **106**, and UE devices **116** are permitted to collectively use solely the frequency and time range specified by PRB **504** to implement wireless communication path **1**.

(47) Method H—Fake Base Station Detection

(48) Method H detects a fake wireless base station. In method H, MRA module **114** and/or MRA module **118** analyze wireless communication signals received at a node in wireless communication network **100**, such as wireless communication signals received by UE device **116** or wireless communication signals received by wireless base station **112**, to determine angle of incidence of the received wireless communication signals. For example, MRA module **114** and/or MRA module **118** may determine angle of incidence of wireless communication signals received by UE device **116**. Reception of a plurality of wireless communication signals at substantially different angles of incidence indicates that the sending wireless base station is likely a legitimate wireless base station. In contrast, reception of a plurality of wireless communication signals at substantially common angles of incidence indicates that the sending wireless base station is likely a fake wireless base station. Accordingly, particular embodiments of MRA module **114** and/or MRA module **118** are configured to determine that a particular wireless base station is fake in response to angle of incidence of received wireless communication signals from the wireless base station differing by less than a predetermined threshold value. Additionally, some embodiment of MRA module **114** and/or MRA module **118** are configured to determine that a particular wireless base station is legitimate in response to angle of incidence of received wireless communication signals from the wireless base station differing by more than a predetermined threshold value.

(49) Method I—Dynamic Power Control

(50) In certain embodiments of wireless communication network **100**, amplification of RF repeaters **106**, **108**, and **110** may be adjusted, and method I includes MRA module **114** and/or MRA module **118** adjusting amplification or “power” of one or more to RF repeaters **106**, **108**, and **110** to optimize or more characteristics of wireless communication network **100**. Method I may be incorporated with any of the methods discussed above. For example, method I may be incorporated with method B to help further increase signal to noise ratio of a sum of copies of a downlink or uplink wireless communication signal.

(51) Load Balancing

(52) A conventional smart RF repeater is capable of receiving data from a donor wireless communication node via a first wireless communication beam and forwarding the data to a child using a second wireless communication beam, in a first slot. The conventional smart RF repeater may be configured such that there is a semi-static linkage between the first wireless communication beam and the second wireless communication beam, and the second wireless communication beam may be dependent on the first slot. Accordingly, a conventional smart repeater is ill-suited for handling aggregated data for one or more target wireless communication nodes over different wireless communication beams. For example, assume a hypothetical scenario where a conventional smart RF repeater receives the following wireless communication signals from a donor: (a) a first wireless communication beam including data **1** for forwarding to a first UE device (UE **1**) and (b) a second wireless communication beam including data **2** for forwarding to a second UE device (UE **2**). A conventional smart RF repeater configured with a semi-static linkage between wireless communication beams would be incapable of determining how to forward data **1** and data **2** to its intended UE device.

(53) Applicant has developed new systems and methods for load balancing which at least partially overcome the above discussed drawbacks of conventional smart repeaters. Certain embodiments are configured to associate frequency regions, or bandwidth parts (BWPs), within a given wireless communication beam with respective destination wireless communication nodes, for forwarding

data. As such, a recipient smart RF repeater, or other recipient network element, can determine where to forward data carried by a wireless communication beam at least partially based on a BWP that the data occupies in the wireless communication beam. Accordingly, particular embodiments of the new systems and methods are capable of receiving data for multiple child wireless communication nodes, e.g., UE devices, customer premises equipment (CPE), and other smart RF repeaters, and forwarding the data to the child wireless communication nodes over different respective wireless communication beams.

(54) FIG. 6 is a block diagram of a wireless communication network **600** including an embodiment of the new systems for load balancing. Wireless communication network **600** includes a wireless communication node **602**, a wireless communication node **604**, a wireless communication node **606**, a wireless communication node **608**, a smart RF repeater **610**, a smart RF repeater **612**, and a smart RF repeater **614**. The aforesaid elements of wireless communication network **600** as configured, for example, in a mesh configuration. The number of wireless communication nodes in wireless communication network **600**, as well as the number of smart repeaters in wireless communication network **600**, may vary without departing from the scope hereof.

(55) Each wireless communication node **602**, **604**, **606**, and **608** is configured to transmit and/or receive wireless communication beams. In some embodiments, each wireless communication node **602**, **604**, **606**, and **608** includes one or more of a wireless base station (e.g., a 3GPP wireless base station, a Wi-Fi wireless base station, a satellite wireless base station, a Bluetooth wireless base station, a LoRa wireless base station, a Zigbee wireless base station, a Z-Wave wireless base station, etc), CPE, a mobile telephone, a computer, a set-top device, a data storage device, an IoT device, an entertainment device, a computer networking device, a smartwatch, a wearable device with wireless capability, a medical device, a security device, and a monitoring device. Each wireless communication node **602**, **604**, **606**, and **608** need not have the same configuration.

(56) Each smart RF repeater **610**, **612**, and **614** is configured to receive data via wireless communication beams and forward data carried by the wireless communication beams to child wireless communication nodes according to respective BWPs associated with the data. Smart RF repeaters **610**, **612**, and **614** could be replaced with other devices capable of forwarding wireless communication beams, e.g., IABs, as long as the other devices are capable of forwarding data carried by wireless communication beams to child nodes according to respective BWPs associated with the data. Additionally, one or more of smart repeaters **610**, **612**, and **614** could be combined with another device, such as a wireless base station, CPE, or a UE device.

(57) FIG. 6 depicts an example of operation of wireless communication network **600** in load balancing where data is being transmitted from wireless communication node **602** to other nodes of wireless communication network **600**, such that wireless communication node **602** is acting as a “donor” wireless communication node. It is understood, though, that wireless communication network **600** could operate with other nodes being donor wireless communication nodes, as well as with wireless communication beams taking paths other than those illustrated in the FIG. 6 example.

(58) In the FIG. 6 example, wireless communication node **602** transmits wireless communication beams **616** and **618** to smart RF repeaters **610** and **612**, respectively, such as in different respective time slots. Each of wireless communication beams **616** and **618** is structured, in the frequency domain, into different BWPs, where each BWP encompasses a respective portion of a wireless communication channel bandwidth of the beam. In particular, wireless communication beam **616** is structured into bandwidth parts BWP01, BWP031, and BWP04, which collectively encompass a bandwidth Ch BW01 of a wireless communication channel associated with wireless communication beam **616**, as illustrated in FIG. 6. Each of bandwidth parts BWP01, BWP031, and BWP04 includes data associated with a different respective wireless communication node of wireless communication network **600**. Specifically, bandwidth part BWP01 includes data associated with smart RF repeater **610**, bandwidth part BWP031 includes data associated with smart RF repeater **614**, and bandwidth part BWP04 includes data associated with wireless

communication node **604**. Wireless communication beam **618**, in turn, is structured into bandwidth parts **BWP02**, **BWP032**, and **BWP05**, which collectively encompass a bandwidth Ch **BW02** of a wireless communication channel associated with wireless communication beam **618**, as illustrated in FIG. **6**. While not required, in certain embodiments, bandwidth Ch **BW01** is equal to bandwidth Ch **BW02**. Each of bandwidth parts **BWP02**, **BWP032**, and **BWP05** also includes data associated with a different respective wireless communication node of wireless communication network **600**. Specifically, bandwidth part **BWP02** includes data associated with smart RF repeater **612**, bandwidth part **BWP032** includes data associated with smart RF repeater **614**, and bandwidth part **BWP05** includes data associated with wireless communication node **606**.

(59) Smart RF repeater **610** receives a message associating bandwidth parts of wireless communication beam **616** with destination wireless communication nodes. Such message may be sent, for example, via data associated with bandwidth part **BWP01**, which is destined for smart RF repeater **610**. The message may associate, for example, (a) bandwidth part **BWP031** with smart RF repeater **614** and (b) bandwidth part **BWP04** with wireless communication node **604**. In some embodiments, the message is radio resource control (RRC) message. Smart RF repeater **610** then forwards data associated with each of bandwidth parts **BWP031** and **BWP04** to its intended destination according to destinations specified in the received message. For example, smart RF repeater **610** may receive wireless communication beam **616** in a first time slot, and smart RF repeater **610** may then forward data associated with bandwidth part **BWP04** to wireless communication node **604** in a second time slot via a wireless communication beam **620**, and smart RF repeater **610** may forward data associated with bandwidth part **BWP031** to smart RF repeater **614** in a third time slot via a wireless communication beam **622**.

(60) Smart RF repeater **612** receives a message associating bandwidth parts of wireless communication beam **618** with destination wireless communication nodes. Such message may be sent, for example, via data associated with bandwidth part **BWP02**, which is destined for smart RF repeater **612**. The message may associate, for example, (a) bandwidth part **BWP032** with smart RF repeater **614** and (b) bandwidth part **BWP05** with wireless communication node **606**. In some embodiments, the message is a RRC message. Smart RF repeater **610** then forwards data associated with each of bandwidth part **BWP032** and bandwidth part **BWP05** to its intended destination according to destinations specified in the received message. For example, smart RF repeater **612** may receive wireless communication beam **618** in a fourth time slot, and smart RF repeater **612** may then forward data associated with bandwidth part **BWP05** to wireless communication node **606** in a fifth time slot via a wireless communication beam **624**, and smart RF repeater **612** may route data associated with bandwidth part **BWP032** to smart RF repeater **614** in a sixth time slot via a wireless communication beam **626**.

(61) Smart RF repeater **614** aggregates (a) data associated with bandwidth part **BWP031** received via wireless communication beam **622**, and (b) data associated with bandwidth part **BWP032** received via wireless communication beam **626**, to yield aggregated data associated with bandwidth part **BWP03**, as illustrated in FIG. **6**. In some embodiments, bandwidth part **BWP03** has a bandwidth Ch **BW03** which is greater than either of bandwidth Ch **BW01** or Ch **BW02**. In some embodiments, smart RF repeater **614** receives each of wireless communication beams **622** and **626** in different respective time slots. Smart repeater **614** then forwards aggregated data associated with bandwidth part **BWP03** to wireless communication node **608** via a wireless communication beam **628**. However, in some alternate embodiments, smart RF repeater **614** is the final destination of aggregated data associated with bandwidth part **BWP03**. Additionally, in some alternate embodiments, smart repeater **614** is replaced with another device, such as CPE, a UE device, or a wireless base station, that is capable of aggregating data associated with of bandwidth part **BWP031** and bandwidth part **BWP032** to yield aggregated data associated with bandwidth part **BWP03**.

(62) In some embodiments, each smart RF repeater **610**, **612**, and **614** forwards received data that

is destined for another wireless communication node without processing the data. Accordingly, in particular embodiments, error management messages, e.g., error correction messages and/or error control messages, are transported from an origination wireless communication node to a destination wireless communication node via one or more smart RF repeaters **610**, **612**, **614** without processing the error correction messages, thereby achieving end-to-end error correction and/or error control. Examples of such error correction messages include, but are not limited to, hybrid automatic repeat request (HARQ) messages, such as HARQ ACK messages and/or HARQ NACK messages. Additionally, some embodiments of wireless communication network **600** are configured to retransmit a data structure, such as a transport block (TB), via an alternate path in the event that the data structure, as received by the destination node, includes one or more errors.

(63) For instance, FIG. 7 is a block diagram of a wireless communication network **700**, which is one embodiment of wireless communication network **600** configured to retransmit a data structure via an alternate path in response to the data structure including errors. Wireless communication nodes **604**, **606**, and **608** of FIG. 6 are not shown in FIG. 7 for illustrative clarity. FIG. 7 depicts an example where a transport block included in bandwidth part BWP**031** includes an error. In particular, wireless communication node **602** transmits a wireless communication beam **702** to smart RF repeater **610**, where wireless communication beam **702** includes bandwidth parts BWP**01**, BWP**031**, and BWP**04**. Smart RF repeater **610** then forwards data associated with bandwidth part BWP**031** to smart RF repeater **614** via a wireless communication beam **704**. Bandwidth part BWP**031** includes a plurality of transport blocks, although only one transport block, i.e., transport block TBx**01**, is shown in FIG. 7 for illustrative clarity. Each transport block includes a plurality of coding block groups (CBGs) (not shown), and each coding block group includes a plurality of coding blocks (CBs) (not shown).

(64) Smart RF repeater **614** processes each coding block of each coding block group in each transport block, to determine if there are any errors. Additionally, smart RF repeater **614** sends a HARQ ACK message to wireless communication node **602** via smart RF repeater **610** and/or **612** in response to each data unit that is successfully received by smart repeater **614**, and smart RF repeater **614** sends a HARQ NACK message to wireless communication node **602** via smart RF repeater **610** and/or **612** in response to each data unit that is not successfully received by smart repeater **614**. As such, wireless communication network **700** supports end-to-end HARQ between wireless communication nodes **602** and **604** and across one or more smart RF repeaters.

(65) In the FIG. 7 example, smart repeater **614** determines that one coding block in a first copy **705** of transport block TBx**01** includes an error. Specifically, coding block CB-**031**-x**02** includes an error as symbolically shown in FIG. 7 by the coding block being filled with cross-hatching, while coding blocks CB-**031**-x**01** and CB-**031**-x**03** do not include errors, as symbolically shown in FIG. 7 by the coding block being filled with light shading. Smart repeater **614** then sends a message to wireless communication node **602** advising of the error in transport block TBx**01**, within a prescribed time interval associated with coding block CB-**031**-x**02**.

(66) Wireless communication node **602** retransmits transport block TBx**01** to smart RF repeater **614** via a different path than that used to originally send transport block TBx**01** to smart RF repeater **614**, in response to receiving the message indicating the error in transport block TBx**01**. In particular, wireless communication node **602** transmits a wireless communication beam **706** to smart RF repeater **612**, with transport block TBx**01** appended to the end of bandwidth part BWP**0321**. Smart RF repeater **612** then forwards data associated with BWP**0321** to smart RF repeater **614** via a wireless communication beam **704**, such that smart RF repeater **614** receives a second copy **710** of transport block TBx**01** via bandwidth part BWP**0321**.

(67) Coding blocks CB-**031**-x**02** and CB-**031**-x**03** in second copy **710** of transport block TBx**01** are error free. However, coding block CB-**031**-x**01** in second copy **710** of transport block TBx**01** includes an error. Nevertheless, smart RF repeater **614** has received error-free copies of each coding blocks CB-**031**-x**01**, CB-**031**-x**02**, and CB-**031**-x**03**, between first copy **705** and second copy

710 of transport block TBx01. Accordingly, smart RF repeater 614 generates a combined copy 712 of transport block TBx01 from error-free coding blocks of each of first copy 705 of TBx01 and second copy of TBx01, using, for example, a chase combining process. While the FIG. 7 example illustrates combined copy 712 transport block of TBx01 being formed from only two copies of the transport block, combined copy 712 of transport block of TBx01 could be formed using additional copies of the transport block, such as by executing a chase combining process a plurality of times.

(68) Combinations of Features

(69) Features described above may be combined in various ways without departing from the scope hereof. The following examples illustrate some possible combinations.

(70) (A1) A method for multiple repeater aggregation includes (1) receiving, at a first node of a wireless communication network, wireless communication signals from a second node of the wireless communication network via a plurality of wireless communication paths, each of the plurality of wireless communication paths including a respective radio frequency (RF) repeater, and (2) selecting one of the plurality of wireless communication paths for use in transferring data between the first node of the wireless communication network and the second node of the wireless communication network, at least partially based on measured conditions of the plurality of wireless communication paths.

(71) (A2) The method denoted as (A1) may further include causing data to be transferred between the first node of the wireless communication network and the second node of the wireless communication network via the selected one of the plurality of wireless communication paths, without coordinating with the respective RF repeaters of the plurality of wireless communication paths.

(72) (A3) In either one of the methods denoted as (A1) and (A2), the first node of the wireless communication network may include a user equipment (UE) device, and the second node of the wireless communication network may include a wireless base station.

(73) (A4) In either one of the methods denoted as (A1) and (A2), the first node of the wireless communication network may include a wireless base station, and the second node of the wireless communication network may include a user equipment (UE) device.

(74) (A5) In any one of the methods denoted as (A1) through (A4), the measured conditions of the plurality of wireless communication paths may include one or more of a reference signal received power (RSRP) of each wireless communication path of the plurality of wireless communication paths, a reference signal received quality (RSRQ) of each wireless communication path of the plurality of wireless communication paths, a time advance of each wireless communication path of the plurality of wireless communication paths, a cell identifier (ID) of each wireless communication path of the plurality of wireless communication paths, a wireless communication beam identifier (ID) of each wireless communication path of the plurality of wireless communication paths, and a percentage of wireless channel utilization of each wireless communication path of the plurality of wireless communication paths.

(75) (A6) In any one of the methods denoted as (A1) through (A5), selecting one of the plurality of wireless communication paths to use for transferring data between the first node of the wireless communication network and the second node of the wireless communication network may include selecting one or more wireless communication beams of the selected one of the plurality of wireless communication paths.

(76) (B1) A method for multiple repeater aggregation includes (1) receiving a plurality of copies of a wireless communication signal, each copy of the wireless communication signal being received via a respective wireless communication path of a plurality of wireless communication paths, each of the plurality of wireless communication paths including a respective radio frequency (RF) repeater, and (2) combining the plurality of copies of the wireless communication signal to generate a combined wireless communication signal.

(77) (B2) In the method denoted as (B1), combining the plurality of copies of the wireless

communication signal to generate the wireless communication signal may include soft combining the plurality of copies of the wireless communication signal.

(78) (B3) In either one of the methods denoted as (B1) and (B2), the steps of receiving and combining may be performed at a user equipment device.

(79) (B4) In either one of the methods denoted as (B1) and (B2), the steps of receiving and combining may be performed at a wireless base station.

(80) (B5) In either one of the methods denoted as (B1) through (B4), the wireless communication signal may include one or more of downlink data and uplink data.

(81) (B6) Any one of the methods denoted as (B1) through (B5) may further include receiving each copy of the wireless communication signal via a respective wireless communication beam.

(82) (C1) A method for multiple repeater aggregation includes (1) receiving, at a first node of a wireless communication network, a plurality of wireless communication signals from a second node of the wireless communication network via a plurality of wireless communication paths, each of the plurality of wireless communication signals representing different data, and each of the plurality of wireless communication paths including a respective radio frequency (RF) repeater, and (2) aggregating the received plurality of wireless communication signals at the first node.

(83) (C2) The method denoted as (C1) may further include causing data to be transferred between the first node of the wireless communication network and the second node of the wireless communication network via the plurality of wireless communication paths, without coordinating with the respective RF repeaters of the plurality of wireless communication paths.

(84) (C3) In either one of the methods denoted as (C1) and (C2), the first node of the wireless communication network may include a wireless base station, and the second node of the wireless communication network may include a user equipment (UE) device.

(85) (C4) In either one of the methods denoted as (C1) and (C2), the first node of the wireless communication network may include a user equipment (UE) device, and the second node of the wireless communication network may include a wireless base station.

(86) (C5) In any one of the methods denoted as (C1) through (C4), at least two wireless communication signals of the plurality of wireless communication signals may be part of a common wireless communication beam.

(87) (C6) In any one of the methods denoted as (C1) through (C5), at least two wireless communication signals of the plurality of wireless communication signals may be within different respective frequencies ranges and/or may be within different respective time ranges.

(88) (D1) A method for load balancing, includes receiving, from a donor wireless communication node, a message, the message associating (a) a first bandwidth part with a first destination wireless communication node, and (b) a second bandwidth part with a second destination wireless communication node.

(89) (D2) The method denoted as (D1) may further include (1) receiving, in a first time slot, data associated with the first bandwidth part, and (2) forwarding the data associated with the first bandwidth part to the first destination wireless communication node, in a second time slot.

(90) (D3) The method denoted as (D2) may further include (1) receiving, in the first time slot, data associated with the second bandwidth part, and (2) forwarding the data associated with the second bandwidth part to the second destination wireless communication node, in a third time slot.

(91) (D4) In any one of the methods denoted as (D1) through (D3), the donor wireless communication node may be a wireless base station.

(92) (E1) A method for load balancing includes (1) receiving, via a first wireless communication beam, data associated with a first bandwidth part of the first wireless communication beam, (2) receiving, via a second wireless communication beam, data associated with a first bandwidth part of the second wireless communication beam, and (3) aggregating (a) the data associated with the first bandwidth part of the first wireless communication beam and (b) the data associated with the first bandwidth part of the second wireless communication beam.

(93) (E2) The method denoted as (E1) may further include receiving the data associated with the first bandwidth part of the first wireless communication beam and the data associated with the first bandwidth part of the second wireless communication beam in different respective time slots.

(94) (E3) Either one of the methods denoted as (E1) and (E2) may further include (1) receiving the data associated with the first bandwidth part of the first wireless communication beam from a first radio frequency (RF) repeater, and (2) receiving the data associated with the first bandwidth part of the second wireless communication beam from a second radio frequency (RF) repeater.

(95) (F1) A method for end-to-end error management in a wireless communication network includes (1) receiving, at a first wireless communication node, data from a second wireless communication node via a first radio-frequency (RF) repeater, and (2) sending one or more error management messages associated with the data from the first wireless communication node to the second wireless communication node via one or more RF repeaters.

(96) (F2) In the method denoted as (F1), the one or more error management messages may include one or more hybrid automatic repeat request (HARQ) messages.

(97) (F3) In either one of the methods denoted as (F1) and (F2), the data received via the first RF repeater may include one or more errors, and the method may further include receiving a copy of the data via a second RF repeater.

(98) (F4) The method denoted as (F3) may further include generating a combined copy of the data from (a) the data received via the first RF repeater and (b) the copy of the data received via the second RF repeater.

(99) (F5) In the method denoted as (F4), generating the combined copy of the data may include using a chase combining process.

(100) Changes may be made in the above methods, devices, and systems without departing from the scope hereof. It should thus be noted that the matter contained in the above description and shown in the accompanying drawings should be interpreted as illustrative and not in a limiting sense. The following claims are intended to cover generic and specific features described herein, as well as all statements of the scope of the present method and system, which as a matter of language, might be said to fall therebetween.

Claims

1. A method for multiple repeater aggregation, comprising: receiving, at a first node of a wireless communication network, wireless communication signals from a second node of the wireless communication network via a plurality of wireless communication paths, each of the plurality of wireless communication paths including a respective radio frequency (RF) repeater; and selecting one of the plurality of wireless communication paths for use in transferring data between the first node of the wireless communication network and the second node of the wireless communication network, at least partially based on measured conditions of the plurality of wireless communication paths, wherein the measured conditions of the plurality of wireless communication paths include one or more of a reference signal received power (RSRP) of each wireless communication path of the plurality of wireless communication paths, a reference signal received quality (RSRQ) of each wireless communication path of the plurality of wireless communication paths, a time advance of each wireless communication path of the plurality of wireless communication paths, a cell identifier (ID) of each wireless communication path of the plurality of wireless communication paths, a wireless communication beam identifier (ID) of each wireless communication path of the plurality of wireless communication paths, and a percentage of wireless channel utilization of each wireless communication path of the plurality of wireless communication paths.
2. The method of claim 1, further comprising causing data to be transferred between the first node of the wireless communication network and the second node of the wireless communication network via the selected one of the plurality of wireless communication paths, without coordinating

with the respective RF repeaters of the plurality of wireless communication paths.

3. The method of claim 1, wherein: the first node of the wireless communication network includes a user equipment (UE) device; and the second node of the wireless communication network includes a wireless base station.

4. The method of claim 1, wherein: the first node of the wireless communication network includes a wireless base station; and the second node of the wireless communication network includes a user equipment (UE) device.

5. The method of claim 1, wherein selecting one of the plurality of wireless communication paths for use in transferring data between the first node of the wireless communication network and the second node of the wireless communication network comprises selecting one or more wireless communication beams of the selected one of the plurality of wireless communication paths.

6. A method for multiple repeater aggregation, comprising: receiving a plurality of copies of a wireless communication signal, each copy of the wireless communication signal being received via a respective wireless communication path of a plurality of wireless communication paths, each of the plurality of wireless communication paths including a respective radio frequency (RF) repeater; combining the plurality of copies of the wireless communication signal to generate a combined wireless communication signal; and selecting one of the plurality of wireless communication paths for use in transferring additional data between a first node of a wireless communication network and a second node of the wireless communication network, at least partially based on measured conditions of the plurality of wireless communication paths, wherein the measured conditions of the plurality of wireless communication paths include one or more of a reference signal received power (RSRP) of each wireless communication path of the plurality of wireless communication paths, a reference signal received quality (RSRQ) of each wireless communication path of the plurality of wireless communication paths, a time advance of each wireless communication path of the plurality of wireless communication paths, a cell identifier (ID) of each wireless communication path of the plurality of wireless communication paths, a wireless communication beam identifier (ID) of each wireless communication path of the plurality of wireless communication paths, and a percentage of wireless channel utilization of each wireless communication path of the plurality of wireless communication paths.

7. The method of claim 6, wherein combining the plurality of copies of the wireless communication signal to generate the combined wireless communication signal comprises soft combining the plurality of copies of the wireless communication signal.

8. The method of claim 6, wherein the steps of receiving and combining are performed at a user equipment device.

9. The method of claim 6, wherein the steps of receiving and combining are performed at a wireless base station.

10. The method of claim 6, wherein the wireless communication signal includes one or more of downlink data and uplink data.

11. The method of claim 6, further comprising receiving each copy of the wireless communication signal via a respective wireless communication beam.

12. A method for multiple repeater aggregation, comprising: receiving, at a first node of a wireless communication network, a plurality of wireless communication signals from a second node of the wireless communication network via a plurality of wireless communication paths, each of the plurality of wireless communication signals representing different data, and each of the plurality of wireless communication paths including a respective radio frequency (RF) repeater; aggregating the received plurality of wireless communication signals at the first node; and selecting one of the plurality of wireless communication paths for use in transferring additional data between the first node of the wireless communication network and the second node of the wireless communication network, at least partially based on measured conditions of the plurality of wireless communication paths, wherein the measured conditions of the plurality of wireless communication paths include

one or more of a reference signal received power (RSRP) of each wireless communication path of the plurality of wireless communication paths, a reference signal received quality (RSRQ) of each wireless communication path of the plurality of wireless communication paths, a time advance of each wireless communication path of the plurality of wireless communication paths, a cell identifier (ID) of each wireless communication path of the plurality of wireless communication paths, a wireless communication beam identifier (ID) of each wireless communication path of the plurality of wireless communication paths, and a percentage of wireless channel utilization of each wireless communication path of the plurality of wireless communication paths.

13. The method of claim 12, further comprising causing data to be transferred between the first node of the wireless communication network and the second node of the wireless communication network via the plurality of wireless communication paths, without coordinating with the respective RF repeaters of the plurality of wireless communication paths.

14. The method of claim 12, wherein: the first node of the wireless communication network includes a user equipment (UE) device; and the second node of the wireless communication network includes a wireless base station.

15. The method of claim 12, wherein: the first node of the wireless communication network includes a wireless base station; and the second node of the wireless communication network includes a user equipment (UE) device.

16. The method of claim 12, wherein at least two wireless communication signals of the plurality of wireless communication signals are part of a common wireless communication beam.

17. The method of claim 12, wherein at least two wireless communication signals of the plurality of wireless communication signals are within different respective frequencies ranges and/or are within different respective time ranges.
